Lecture Lab example notes

Squares and derivatives

For a real number x, its square is x multiplied by x.

For x = 3, the value of x squared is 9.

The derivative of f(x) = x squared is f'(x) = 2x.

At x = 3, the derivative f'(3) is 6.

Lecture Lab example notes

Bernoulli random variables

A Bernoulli random variable X takes only the values 0 and 1.

If the success probability is p, then P(X = 1) = p.

Its expectation is E[X] = p and its variance is Var(X) = p(1 - p).

For p = 0.25, the expectation is 0.25 and the variance is 0.1875.